

3.2

$$\langle f, g \rangle = \int_1^5 f(x)g(x)dx$$

a) označme $\vec{v}_1 = 1$ $\vec{v}_2 = x$ $\vec{v}_3 = x^2$

$B = (\vec{u}_1, \vec{u}_2, \vec{u}_3)$ ortonormalna baza

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{2}$$

$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|}$ kde $\vec{w}_2$ je ortogonalna projekcia $\vec{v}_2$ na $\text{span}\{\vec{u}_1\}$

$$\vec{w}_2 = \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1 = \int_1^5 \frac{1}{2} x dx \cdot \frac{1}{2} = 3$$

$$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\frac{8}{\sqrt{3}}} = \frac{\sqrt{3}}{8}(x-3)$$

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} \quad \vec{w}_3 = \langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2 = \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{8} (x-3)^2 x dx \vec{u}_2$$

$$= \frac{27}{16} x + \frac{37}{16}$$

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} = \frac{x^2 - \frac{27}{16}x - \frac{37}{16}}{\sqrt{\int_1^5 \left(x^2 - \frac{27}{16}x - \frac{37}{16}\right)^2 dx}} = \frac{3\sqrt{5}}{16} \left(x^2 - \frac{27}{16}x - \frac{37}{16}\right)$$

ortonormalna baza je:

$$B = (\vec{u}_1, \vec{u}_2, \vec{u}_3) = \left(\left(\frac{1}{2}\right), \left(\frac{\sqrt{3}}{8}(x-3)\right), \left(\frac{3\sqrt{5}}{16} \left(x^2 - \frac{27}{16}x - \frac{37}{16}\right)\right) \right)$$

$\rightarrow$ b) máme vyřátané $\vec{u}_1 = \frac{1}{2}$ $\vec{u}_2 = \frac{\sqrt{3}}{8}(x-3)$ $\vec{u}_3 = \frac{3\sqrt{5}}{16}\left(x^2 - \frac{27}{16}x - \frac{37}{16}\right)$
 prvky ortonormalní báze

$\times$ d také ortogonální projekce $\sin x$ na $\text{span}\{1, x, x^2\} = \text{span}\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ je

$$\vec{S} = \langle \vec{u}_1, \sin x \rangle \vec{u}_1 + \langle \vec{u}_2, \sin x \rangle \vec{u}_2 + \langle \vec{u}_3, \sin x \rangle \vec{u}_3$$

co po vyřátání a úpravách dá

$$= -0,0883x^2 + 0,2081x + 0,4947$$

e)

$$\| \sin x - (ax^2 + bx + c) \| = \sqrt{\int_0^5 (\sin x - (-0,0883x^2 + 0,2081x + 0,4947))^2 dx} = 0,86768$$